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Final Exam

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Wintersemester 2021
40 Points

Mathematics for Machine Learning

1. Linear Algebra

a) Let $A \in \mathbb{R}^{2\times 2}$ be a matrix representation of a linear map with respect to the canonical basis. Find the matrix B representing the same transformation with respect to arbitrary basis vectors $v_1, v_2 \in \mathbb{R}^2$. (3)

b) Suppose $S : V \to V$ is a linear map on a vector space V. Define $\langle \cdot, \cdot \rangle_1$ by

$$\langle u, v \rangle_1 = \langle Su, Sv \rangle$$

for $u, v \in V$. Show that $\langle \cdot, \cdot \rangle_1$ is an inner product on V if S is injective. (3)

c) Let $A \in \mathbb{R}^{n\times n}$ be symmetric. The Rayleigh-quotient is defined as

$$R_A(x) = \frac{x^T A x}{x^T x}.$$

Show that

$$\max_{x \neq 0} R_A(x) = \lambda_{\max},$$

where $\lambda_{\max}$ is the largest eigenvalue of A. (4)

2. Calculus

a) What is the directional derivative of $f(x, y) = x^2 + xy$ in the direction $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ at point(1, 1)? (2)

b) We want to find the point(s) in the curve $y = 1.5 - x^2$ closest to the origin using the squared Euclidean distance as a metric. What objective needs to be minimized? What is the constraint function? Plot the curve, gradients of the constraint, and the gradients and levelsets of the objective function (in 2D), as well as the constrained minimizers. Describe the necessary and sufficient conditions which need to be satisfied for a point to be a constrained minimizer. (4)

c) Let $f : \mathbb{R}^3 \to \mathbb{R}^3$ be a function with $f(x) = (f_1(x), f_2(x), f_3(x))^T$, where $f_i : \mathbb{R}^3 \mapsto \mathbb{R}, i \in \{1, 2, 3\}$. What is the Jacobian matrix of f? Let $p \in \mathbb{R}^3$ be an arbitrary point. Suppose we want to minimize the objective $\min_x \|p - f(x)\|_2^2$. Starting from an initial guess x_0, we proceed by $x_{k+1} = x_k + \Delta x_k$. Instead of following gradient descent, we will find a step direction Δx_k differently. Make use of the first-order expansion of $f(x_k + \Delta x_k)$, and find the Δx_k that minimizes the approximation of the objective. (4)

3. Optimization

a) Consider an optimization problem $\min_{x\in\Omega} f(x)$, where f is a convex function and Ω is a convex set. Prove that if x is a local minimum, then it is also a global minimum: $f(y) > f(x), \forall y \in \Omega$. (Hint: Prove by contradiction.) (3)

b) In gradient descent with line search, we search for the optimal step size α_k in each update step: $x_{k+1} = x_k + \alpha_k d_k$, where d_k is the descent direction. Let $\{x_k\}$ be a sequence generated by steepest descent with line search. Show that $x_{k+2} - x_{k+1}$ is orthogonal to $x_{k+1} - x_k$ for every k. (Hint: $\frac{df}{d\alpha} f(x + \alpha h) = \langle h, \nabla f(x + \alpha h) \rangle$) (3)

c) Formulate an optimization problem and use the KKT conditions to find the closest point to $(0, 0)$ in the set defined by (4)

$$M = \{x \in \mathbb{R}^2 : x_1 + x_2 \geq 4, 2x_1 + x_2 \geq 5\}.$$

4. Probability

a) If one tosses a fair coin twice. Let X be defined as the number of heads observed. Find the probability mass function P_X. (1)

b) Suppose a coin has a probability p for showing head. If one tosses this coin twice, let X denote the result of first toss and Y denote the result of second toss (1 for head and 0 otherwise). Find the expectation $\mathbb{E}[XY]$. (2)

c) Assume one has three bags that each contain 100 marbles: Bag 1 has 75 red and 25 blue marbles, Bag 2 has 60 red and 40 blue marbles, Bag 3 has 45 red and 55 blue marbles. If one randomly picks a marble from an unknown bag and it turns out to be a red marble, what is the probability that this marble is picked from Bag 1? (3)

d) Consider N data-points of the form $\{x_n, y_n\}_{n=1}^N$ where $x_n \in \mathbb{R}^D$ and $y_n \in \mathbb{R}$. We refer to x as the input features and to y as the targets. Assume that the data arises from a process $y = x^T\Theta + \epsilon$ where $\Theta \in \mathbb{R}^D$. The noise ϵ comes from a standard normal distribution and is i.i.d. $\epsilon \sim \mathcal{N}(0, \sigma^2)$

Let $X \in \mathbb{R}^{N \times D}$ denote all the input features stacked together in a design matrix and $Y \in \mathbb{R}^N$ denote the N corresponding targets stacked together. Using the i.i.d assumption of the noise, write the conditional probability of the N data-points $P(Y|X;\Theta)$. Then show that the maximum likelihood estimate of Θ is

$$\Theta = (X^T X)^{-1} X^T Y$$

(4)